Arabic Sentiment Stability Under Paraphrasing

Comparing Gemini, GPT-OSS, and AraBERT

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Motivation

Arabic sentiment analysis is challenging because Arabic social media includes:

- dialectal variation,
- informal spelling,
- sarcasm and irony,
- short and noisy text,
- neutral news content with emotional topics.

Accuracy alone does not show whether a model stays stable when the same meaning is expressed differently.

Research Question

Main question:

Do Arabic sentiment models keep the same prediction when Arabic tweets are paraphrased?

We study:

- Which model is most stable?
- Which sentiment classes are most fragile?
- Do dialect and sarcasm affect stability?
- Can few-shot prompting improve robustness?

Datasets

We combined two public Arabic Twitter datasets:

- **ASTD**: Arabic Sentiment Tweets Dataset
- **ArSarcasm**: Arabic sarcasm and sentiment dataset

Processed dataset:

- 20,241 tweets
- Unified labels: positive, negative, neutral

Experiment sample:

- 150 tweets
- 50 positive, 50 negative, 50 neutral

Dataset Overview

Item	Value
Total tweets	20,241
ASTD tweets	9,694
ArSarcasm tweets	10,547
Positive	2,455
Negative	5,171
Neutral	12,615
Number of dialect values	6

Methodology

Pipeline:

1. Sample 150 balanced tweets.
2. Generate one paraphrase per tweet using **Gemini 2.5 Flash**.
3. Classify original and paraphrased tweets using:
    - Gemini 2.5 Flash
    - GPT-OSS-20B via Groq
    - AraBERT-ArSAS
4. Compare predictions before and after paraphrasing.

Evaluation Metrics

- **Accuracy**: prediction compared with gold label.
- **Macro-F1**: balanced performance across classes.
- **Consistency rate**: percentage of unchanged predictions.
- **Flip rate**: percentage of changed predictions.

$$Consistency = \frac{\#\ \text{unchanged predictions}}{\#\ \text{original-paraphrase pairs}}$$

$$FlipRate = 1 - Consistency$$

Consistency & Flip

Consistency rate: The percentage of paraphrased tweets where the model gives the same sentiment label as the original tweet.

Flip rate: The percentage of paraphrased tweets where the model changes its sentiment label compared to the original tweet.

Consistency rate

$3 / 4 \times 100 = 75\%$

Flip rate

$1 / 4 \times 100 = 25\%$

Original Arabic tweet	gold_label	Pred Original label	Paraphrased tweet	Pred New label	Result
القمح إلى #أوكرانيا يهدد واردات #تفاقم أزمة الدول العربية	Neutral	negative	تصاعد الأزمة في أوكرانيا يمثل تهديداً لوصول القمح إلى الدول العربية.	negative	Consistent
شكر ل عآدل عبداالله ع البنتي و الطرد _كلمة# 😂😂 اللي حصله ، أصلاً الطرد احلى	Positive	positive	حابين نشكر عادل عبدالله على البنتي اللي أصلاً، أخذه. وعلى الطرد اللي حصله كمان الطرد كان أحلى!	positive	Consistent
#الله يبركلك يا دكتور ابو الفتوح	Positive	positive	ربنا يبارك فيك يا دكتور ابو الفتوح	positive	Consistent
😂😂 الارجنتين بدون ميسي مضحكه	Negative	positive	الأرجنتين بلا ميسي مسخرة.	negative	Flip

Models

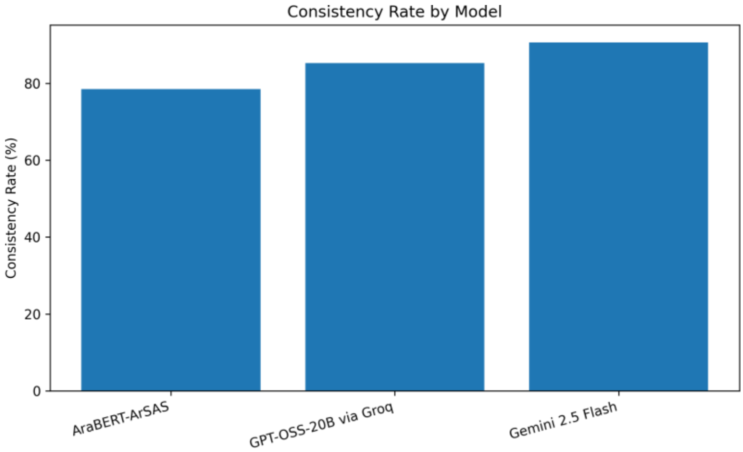
Model	Role
Gemini 2.5 Flash	LLM for paraphrasing and sentiment classification
GPT-OSS-20B via Groq	Open GPT-style LLM for classification
AraBERT-ArSAS	Arabic transformer sentiment classifier

Baseline Results

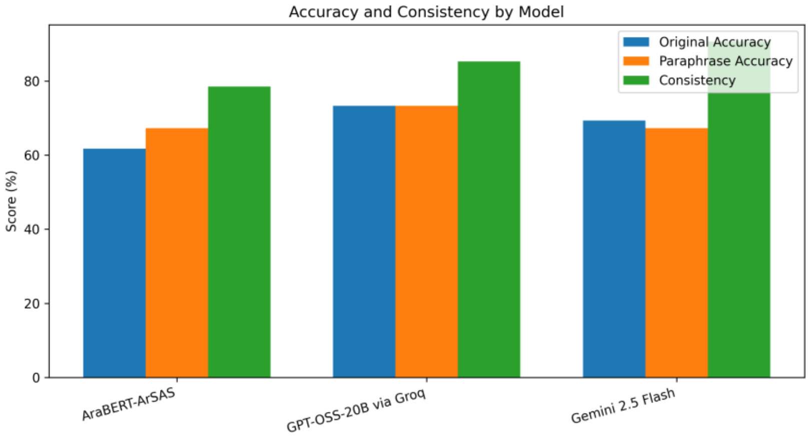
Model	Orig. Acc.	Para. Acc.	Orig. F1	Para. F1	Cons.	Flip
AraBERT-ArSAS	61.74%	67.33%	61.68%	66.72%	78.52%	21.48%
GPT-OSS-20B	73.33%	73.33%	72.96%	72.76%	85.33%	14.67%
Gemini 2.5 Flash	69.33%	67.33%	66.63%	65.26%	90.67%	9.33%

Key point: Gemini was the most stable, while GPT-OSS had the strongest accuracy and macro-F1.

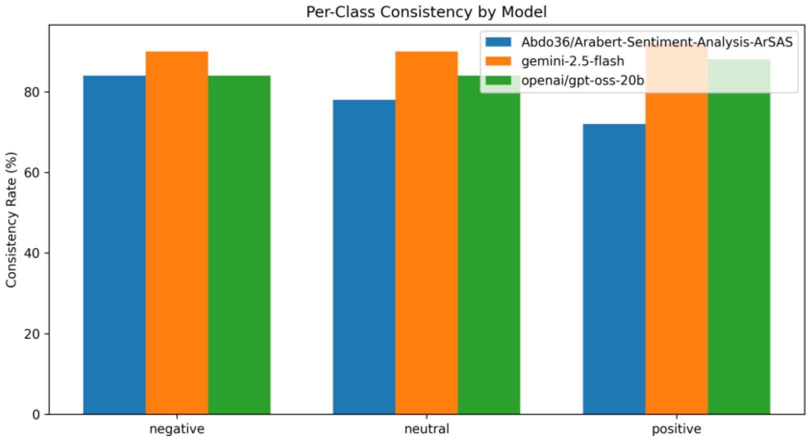
Consistency by Model



Accuracy vs Consistency



Per-Class Consistency



Sample Sentiment Flip Cases

Model	Original Text	Paraphrase	Orig.	Para.	Explanation
Gemini	مصر دولة بس مش أوي	مصر دولة على الورق، لكن في الحقيقة هي مش كده.	neg.	pos.	Sarcasm became less direct after paraphrasing.
Gemini	صورة: كاسياس يحتفل بالانتصار في ديربي مدريد	تُظهر هذه الصورة كاسياس وهو يحتفل بالانتصار في ديربي مدريد.	pos.	neu.	Paraphrase became more descriptive and less emotional.
GPT-OSS	الارجنتين بدون ميسي مضحكة	الأرجنتين بلا ميسي مسخرة.	pos.	neg.	The paraphrase made the criticism clearer.
AraBERT	عجيييييب	خرافي!	pos.	neg.	Short informal expression was misclassified.

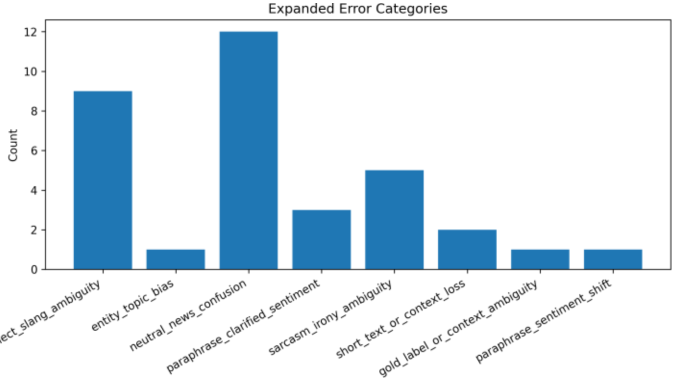
Flips often happen when paraphrasing changes the strength or clarity of sentiment cues.

Expanded Error Analysis: Summary

Error Category	Count	Percentage
Neutral-news confusion	27	39.71%
Dialect/slang ambiguity	17	25.00%
Sarcasm/irony ambiguity	13	19.12%
Paraphrase clarified sentiment	6	8.82%
Short text/context loss	2	2.94%
Entity/topic bias	1	1.47%
Paraphrase sentiment shift	1	1.47%
Gold-label/context ambiguity	1	1.47%

Most flips came from neutral-news confusion, dialect/slang ambiguity, and sarcasm.

Expanded Error Analysis: Chart

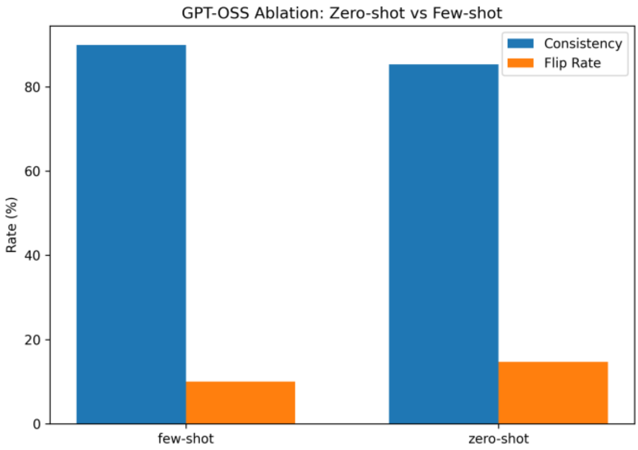


Ablation: Zero-shot vs Few-shot GPT-OSS

Config.	Orig. Acc.	Para. Acc.	Orig. F1	Para. F1	Cons.	Flip
Zero-shot	73.33%	73.33%	72.96%	72.76%	85.33%	14.67%
Few-shot	73.33%	73.33%	73.15%	73.21%	90.00%	10.00%

Few-shot prompting improved consistency from **85.33%** to **90.00%**, while accuracy stayed the same.

Ablation: Consistency and Flip Rate



Statistical Significance

Comparison	p-value	Significant?
Gemini vs GPT-OSS zero-shot	0.2005	No
Gemini vs AraBERT	0.0066	Yes
GPT-OSS zero-shot vs few-shot	0.1185	No

- Gemini was significantly more stable than AraBERT.
- Gemini vs GPT-OSS was not statistically significant.
- Few-shot improved GPT-OSS numerically, but not significantly.

Main Findings

- Arabic sentiment predictions are not fully stable under paraphrasing.
- **Gemini** achieved the highest consistency rate.
- **GPT-OSS** achieved the strongest accuracy and macro-F1.
- **AraBERT** had the highest flip rate.
- The most common error types were:
  - neutral-news confusion,
  - dialect/slang ambiguity,
  - sarcasm/irony ambiguity.

Limitations and Future Work

Limitations

- Small sample
- Paraphrases generated by one model.
- limited sizes for some dialect groups.

Future work

- Use larger samples.
- Generate multiple paraphrases per tweet.
- Run dialect-specific experiments.

Conclusion

- Stability and accuracy are not the same.
- Gemini was most stable, but GPT-OSS was most accurate.
- Paraphrasing can reveal hidden model fragility.
- Few-shot prompting can improve stability
- Arabic sentiment evaluation should include robustness metrics.